Read the following scenario and select the most appropriate answer for the questions 1–5:

A university is redesigning its departmental website. The homepage contains a top banner, a navigation menu, a welcome section, and a contact form. The developer wants the page title to appear on the browser tab, important keywords to be included for search engines, and the navigation links to lead users to different pages. The contact form should require students to enter a valid email address before submission.

1. Which HTML tag should be used to define the text shown on the browser tab?
a) <meta>
b) <title>
c) <header>
d) <headings>

2. Which HTML element is most appropriate for grouping the site's main navigation links?
a) <section>
b) <aside>
c) <nav>
d)

3. To provide metadata such as page description or keywords, which tag should be used inside the <head> section?
a) <meta>
b) <footer>
c) <legend>
d)

4. Which attribute in an <a> tag specifies the destination page?
a) src
b) id
c) href
d) value

5. Which input type is most suitable for ensuring that the user enters an email in proper format?
a) text
b) number
c) email
d) file

Read the following scenario and select the most appropriate answer for the questions 6–10:

A startup company builds a task management web app. The browser sends requests to a web server, the business rules are handled by an application server, and user records are stored in a database server. The company also uses a DNS server so users can access the app using a domain name instead of remembering an IP address. For added security, the company wants all login traffic to be encrypted.

6. Which server is responsible for storing and retrieving user records?
a) Proxy server
b) DNS server
c) Database server
d) Web server

7. Which type of architecture is being used when presentation, logic, and data are handled separately?
a) Peer-to-peer architecture
b) Three-tier architecture
c) Single-tier architecture
d) Ring architecture

8. Which server helps translate a domain name into its corresponding IP address?
a) Application server
b) DNS server
c) Database server
d) Proxy server

9. Which protocol should be used to encrypt login data between the browser and server?
a) HTTP
b) FTP
c) HTTPS
d) SMTP

10. In client-server communication, which machine usually initiates the request for a webpage?
a) Database server
b) Client
c) DNS server
d) Proxy server

Read the following scenario and select the most appropriate answer for the questions 11–15:

An online bookstore displays books in product cards. Each card contains a book cover image, title, price, and a short description. The developer wants all cards to have equal spacing inside their borders, a visible outline when selected, and enough margin between one card and the next. Some descriptions are too long and spill out of their boxes. The developer also wants the card titles to appear bold.

11. Which CSS property controls the space between an element's border and its content?
a) margin
b) padding
c) outline
d) display

12. Which CSS property controls the space outside the border, separating one card from another?
a) border
b) padding
c) margin
d) z-index

13. Which CSS property can be used to show a line around an element without affecting its box size?
a) outline
b) border
c) overflow
d) float

14. Which CSS property should be used when text is overflowing outside a fixed-size description box?
a) position
b) overflow
c) flex-wrap
d) top

15. Which HTML tag is most appropriate for making the card title text semantically strong?
a) <b>
b)
c)
d) <i>

Read the following scenario and select the most appropriate answer for the questions 16–20:

A travel agency webpage contains a search form where users enter destination, travel dates, and number of travelers. When the user clicks the “Search” button, JavaScript checks whether the fields are filled correctly. If the input is valid, the page displays matching tour packages. The developer stores package prices in an array and uses a loop to display each result.

16. Which JavaScript structure is most suitable for running one block of code when input is valid and another when it is not?
a) for loop
b) if...else
c) object literal
d) array method

17. Which JavaScript data structure is best for storing multiple package prices?
a) string
b) boolean
c) array
d) null

18. Which loop is commonly used to process array elements one by one?
a) switch
b) while
c) for
d) ternary

19. Which event is most closely associated with clicking the “Search” button?
a) hover
b) load
c) click
d) resize

20. Which JavaScript keyword is used to declare a variable whose value may change later?
a) const
b) let
c) class
d) static

Read the following scenario and select the most appropriate answer for the questions 21–25:

A news portal has a header at the top, a floating weather widget on the right, and a “Subscribe Now” button that should stay visible even when the user scrolls down the page. A notification badge must appear above all other elements. The developer also wants two small labels to sit on the same line while still allowing width and height to be set.

21. Which CSS position value keeps the “Subscribe Now” button visible even during scrolling?
a) static
b) relative
c) fixed
d) absolute

22. Which CSS property determines which element appears on top when elements overlap?
a) display
b) z-index
c) float
d) clear

23. Which display value allows an element to stay on the same line but still accept width and height?
a) block
b) none
c) inline
d) inline-block

24. Which position value places an element relative to its normal starting position?
a) fixed
b) relative
c) static
d) sticky-only

25. Which position value removes an element from normal flow and places it relative to the nearest positioned ancestor?
a) absolute
b) static
c) block
d) float

Read the following scenario and select the most appropriate answer for the questions 26–30:

A food delivery service uses a webpage where customers select meal options, choose toppings, and upload a payment screenshot. The topping choices allow multiple selections, but the delivery type allows only one option: “Regular” or “Express.” The developer also wants a reset button to clear the form and a submit button to send it.

26. Which input type is most suitable when users can select multiple toppings?
a) radio
b) checkbox
c) number
d) email

27. Which input type is most suitable when users must select only one delivery type from two options?
a) checkbox
b) text
c) radio
d) file

28. Which input type should be used for uploading the payment screenshot?
a) image
b) file
c) url
d) hidden

29. Which form control is used to send the completed form data?
a) reset
b) submit
c) radio
d) select

30. Which form control clears all user-entered values back to their defaults?
a) reset
b) file
c) button-only
d) legend

Read the following scenario and select the most appropriate answer for the questions 31–35:

A developer is building a photo gallery page. The images are placed inside a container. Some photos should wrap to the next line when the screen becomes smaller. The developer also wants one old advertisement panel to disappear completely without leaving empty space. A special style should be applied when the user hovers over a thumbnail, and the first letter of each caption should appear larger.

31. Which CSS property is used in a flex container to allow items to move to the next line?
a) float
b) flex-wrap
c) overflow
d) z-index

32. Which CSS value completely hides an element and removes it from the page layout?
a) visibility: hidden
b) display: none
c) opacity: 0
d) position: absolute

33. Which type of selector is used to style an element when the mouse pointer is over it?
a) pseudo-element
b) pseudo-class
c) id selector
d) group selector

34. Which type of selector is used to style the first letter of a caption differently?
a) pseudo-class
b) group selector
c) pseudo-element
d) combinator

35. Which CSS selector would target all elements with class name thumb?
a) #thumb
b) thumb
c) .thumb
d) *thumb

Read the following scenario and select the most appropriate answer for the questions 36–40:

A movie streaming platform tracks a user's watchlist using JavaScript. When a user adds a movie, it is inserted into an array. If they remove the last movie added, the most recent entry is deleted. The developer also stores the currently selected movie inside an object with properties such as title and genre. To update the page, JavaScript creates new list items and removes old ones from the DOM.

36. Which array method removes the last element from an array?
a) push()
b) pop()
c) shift()
d) join()

37. Which array method adds a new movie to the end of the watchlist?
a) pop()
b) push()
c) slice()
d) sort()

38. Which JavaScript structure is most suitable for storing a movie's title and genre together?
a) array
b) loop
c) object
d) string

39. Which DOM method is commonly used to create a new HTML element such as ?
a) document.makeElement()
b) document.createElement()
c) document.appendElement()
d) document.newNode()

40. Which DOM operation is used to remove an existing element from the page?
a) delete style
b) remove the node/element
c) hide with value
d) reset innerHTML only